ONE-PAGE CONCEPT SUMMARY

Inference-time scaling without a verbal scratchpad

Chain-of-thought allocates extra inference compute by decoding tokens and feeding them back. That works, and it is expensive: every intermediate state must pass through a vocabulary, an autoregressive step, and a growing context. **Inference-time scaling via latent computation** spends the same budget inside a recurrent hidden state and emits only the answer. The current reason it matters is that 2024–2026 reasoning systems (o1-class token budgets, Coconut-style continuous thoughts, recurrent-depth LMs, HRM/TRM) all buy accuracy with extra test-time work, but they do not buy it the same way.

Mechanism

BDH-CQ (Engdahl et al., 2026) is a 150M-parameter reasoner in the Dragon Hatchling (BDH) family. Demonstrations of an unseen task update a recurrent associative memory `S_t = U_theta(S_{t-1}, D_t)` with theta frozen. After K examples, a latent workspace is encoded and iterated `H_{r+1} = F_theta(H_r, S_K)`, then decoded. No evaluation-task backward pass, no puzzle identity embedding, no verbal trace. Linear attention is the special case `S_t = S_{t-1} + U_theta(D_t)`. Dimensions of U_theta and F_theta are proprietary; this is a published interface, not a full recipe. BDH itself is not an SSM in the Mamba sense: BDH-GPU is ReLU-low-rank communication plus linear attention, with Hebbian synaptic working memory, ~5% sparse non-negative activations, and reported monosemantic synapses (Kosowski et al., 2025).

What changes

Extra compute is a deeper unrolling of H, not a longer string. Memory stays fixed-size; cost and latency do not inherit decode-length. The trade: the trace is unobservable, superposition can interfere, and composition is not guaranteed just because R increased.

Evidence (labeled)

On the 400-task public ARC-AGI-1 split, HIGH effort is **118/400 = 29.5% pass@2** (Wilson 25.2–34.2%) at a computed **$0.00070/task** (0.85 H200-s at $3/h). pass@1 is 97/400 = 24.25%. Table 5, on a model trained with multiple latent-reasoning levels, reports **LOW 21%, MEDIUM 27%, HIGH 29.5%**, all with zero CoT tokens. A black-box audit (Kinas / Zhong) reproduced 29.5% without weights. This is developer-reported hardware cost plus an independent score check — not an independent reimplementation.

Comparators that actually matter

Luna (Low) is **more accurate** (34.2%) and, on the 4 Aug 2026 listing, **57x costlier** at $0.040 (~11x after the July cut). BDH-CQ's claim is the in-context cost frontier, not a new accuracy record. Snell et al. (2024) is the token-budget citation; MATH/GSM8K curves do not belong on this ARC plane.

	CoT / o1	HRM / TRM	BDH-CQ
Extra compute	decoded tokens	eval-task opt.	latent H
Eval backward pass	no	yes	no
This plane	Luna 34.2% @ $0.040	$1.48 / $1.76	29.5% @ $0.0007

Where it is weaker

ConceptARC strict-task pass@2 is 59.4% while pair accuracy is 77.9%: many tasks are only partly solved. Copy and Order are 2/10; FilledNotFilled and TopBottom2D are 9/10. Dense color maps bind 96/96; color-swap composed with relocation is **0/72**. Generated gravity-and-stacking is 2.9% against flood-fill 68.6%. §6.6 MIN vs STANDARD on the deployed system is 111 vs 118 / 400 ($p = 0.167$): the large Table 5 jump is a train-and-select protocol, not a proven one-checkpoint slider.

Maturity and the limitation to keep

Early BDH pretraining is reported from 1B to 600B with Transformer-like scaling; that is not the 150M ARC system. AWS/SageMaker integration is a deployment partnership, not an independent scientific evaluation. Latent effort only refines an operator the memory already bound. More R will not invent a missing composition. Observability is the other tax: when HIGH fails, there is no chain of thought to read.

Continue: arXiv:2608.09888 (BDH-CQ); arXiv:2509.26507 (BDH); Geiping et al. 2025 (recurrent depth); Hao et al. 2024 (Coconut); Snell et al. 2024.